



Carnegie Mellon University
Master of
Software Engineering

Energy efficiency, interoperability,
portability, scalability, deployability

LG Architecture Training Program

Paulo Merson

Agenda



→ Energy efficiency
Interoperability
Portability, scalability,
deployability

Energy consumption is a concern for...



Cloud provider data centers...



... as well as IoT and mobile devices

Energy efficiency

- QA that gained relevance with
 - Mobile devices, IoT, and battery-powered devices in general
 - Large server farms of cloud providers
- Common tradeoffs are performance, availability, and cost



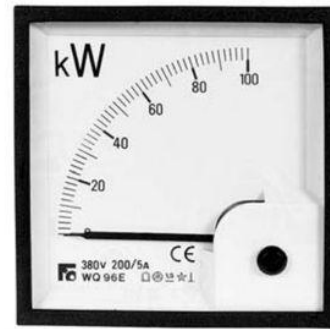
Tactics for energy efficiency

- Metering
- Reduce usage (of energy)
- Reduce computational overhead
- Manage work requests (sampling rate or event arrival)

*The last two are
performance tactics that
reduce energy demand*



Metering



- Essential tactic for monitoring resources
- Individual servers can be measured via amp meters
- In mobile devices, modern batteries have APIs to query and manage battery state

You can't manage what you can't measure

Reduce usage (of energy)

- Achieved through different strategies, such as
 - Reducing the refresh rate of a display
 - Making the display darker (dim screen backlight; dark theme)
 - Stopping or pausing processes that have been inactive for a while; sleep mode after N minutes of inactivity
 - Lower CPU clock rate
- In some cases, mobile apps can save energy by sending computation to the cloud

Agenda

Energy efficiency

➔ Interoperability

Portability, scalability,
deployability

Interoperability*

Interoperability is the ability of two software elements to interact and share information according to an agreed-upon operational semantics
[Brownsword04]

- The software elements
 - may run on the same platform or different platforms
 - should be **syntactically interoperable**: use compatible technologies and exchange data in a standard format
 - should be **semantically interoperable**: agree on the meaning of the operations and data elements used in their interaction

* Called integrability in [Bass21] (was interoperability in previous editions 🤔)

Lack of semantic interoperability example

New York Giants at Green Bay Packers - weather.com - ...

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Sports & Recreation: Weather for NFL Games > **Game Forecast**

Yesterday Today Tomorrow Radar Map Hour-by-Hour

NFL Game Weather



New York Giants AT Green Bay Packers

Sun., Jan. 20, 2008
Time: 5:30 PM CST
Lambeau Field
Green Bay, WI

New York Giants
[Schedule](#)
[Best New York Giants Weather](#)

Green Bay Packers
[Schedule](#)
[Best Green Bay Packers Weather](#)

Right Now at the Stadium



Clear

Temperature: -2°F
Feels like: -22°F
Sunset: 4:44 PM
[Spectator Index: 1](#)
[UV Index: Low](#)

Wind Direction


Wind From the West at 14 mph

Wind Speed


Medium

Game Details

New York Giants at Green Bay Packers - weather.com - ...

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[Schedule](#)
[Best New York Giants Weather](#)

Green Bay Packers
[Schedule](#)
[Best Green Bay Packers Weather](#)

Right Now at the Stadium



Clear

Temperature: -19°F
Feels like: -30°F
Sunset: 4:44 PM
[Spectator Index: 1](#)
[UV Index: Low](#)

Wind Direction


Wind From the West at 23 mph

Wind Speed


Strong

Game Details

<http://www.weather.com/outlook/events/sports/pastweather/29825:20>

Interoperability tactics

- Provide a standardized contract
- Use an intermediary
- Service Oriented Architecture (SOA)

We'll talk
about SOA
in week 4



Provide a standardized contract

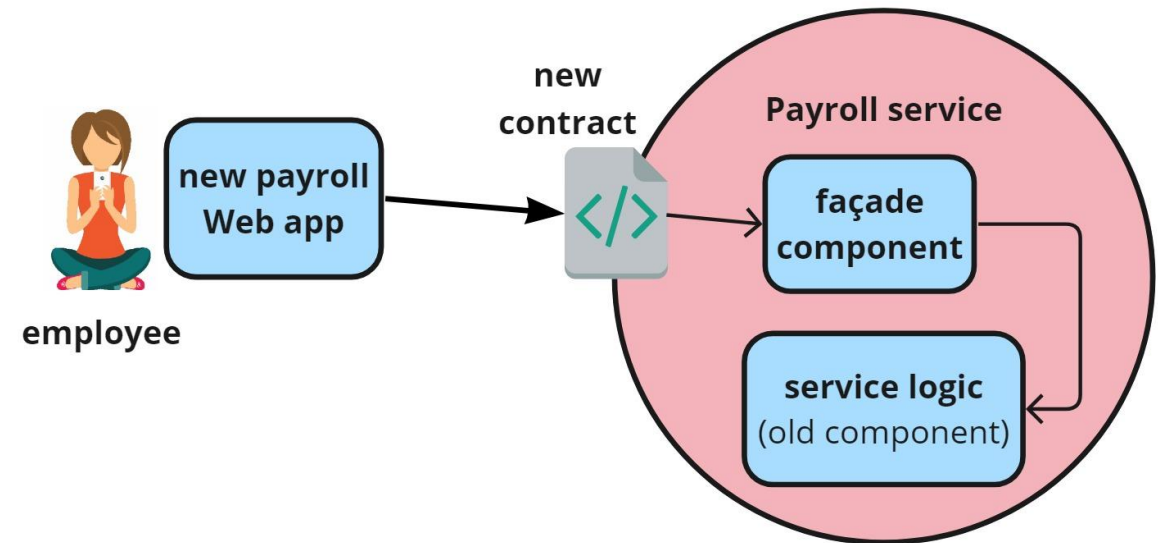
- Standardization includes:
 - Use of API design standards
 - Use of industry standards (e.g., JSON, XML)
 - Standardization of data representation used in input and output messages
- **Contract-first design** (aka API design first [Higginbotham21]) recommends that we define the contract first, and then implement the component



What if the component is already implemented?

Contract-first with existing components

- Contract-first design can be done even if an existing component provides the service logic
- A solution is to add a **façade component** between the contract and the logic to deal with any discrepancy between the new contract and the old logic
- The façade logic can be responsible for
 - data transformation
 - dispatching calls to the correct logic component (there can be 2+)
 - processing data in headers
 - authorization checks



Use an intermediary



- An intermediary handles discrepancies that would otherwise make the two software elements incompatible (non interoperable)
- To enable interoperability the intermediary may perform:
 - Message format conversion, such as JSON to XML
 - Message structure transformation, such as UserAndroid.xsd to User.xsd
 - Protocol bridging, such as converting from HTTP to gRPC
- Examples of patterns that may provide this ability
 - Interceptor [discussed next week]
 - Wrapper [discussed in week 4]
 - API gateway (used in distributed systems and cloud computing)

Agenda

Energy efficiency

Interoperability

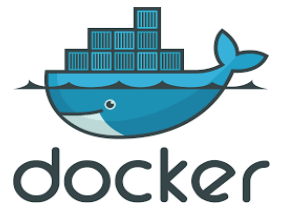
➔ Portability, scalability,
deployability

Portability

- Ability to adapt the system to a different runtime platform
- Tactics:
 - Minimize platform dependencies in the code
 - Example: if you use Oracle database, don't use proprietary Oracle SQL functions
 - Isolate platform-specific code through
 - APIs
 - layer or virtual machine
 - platform-specific wrappers or adapters

Portability is a specific aspect of modifiability

Containers are a form of virtualization that promotes this isolation



We'll talk about containerization in week 3

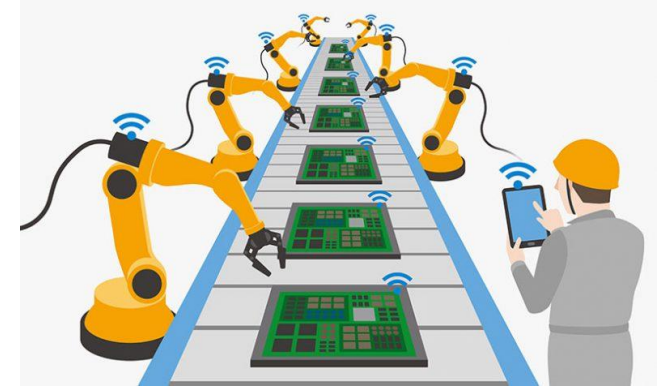
Scalability

- If the load increases (number of requests, users, etc.) the supporting resources should increase accordingly
- The system is scalable if resources increase timely to support the increasing load
- Two types of scalability:
 - **Vertical (scaling up)**: increase the capacity of a machine
 - **Horizontal (scaling out)**: increase the number of machines
- Examples of design strategies that support scalability:
 - Stateless services
 - Introduce concurrency (asynchronous processing)
 - Passive redundancy with load-balancing
 - Serverless architecture

Elasticity requires increasing and decreasing capacity according to the load



Deployability



- Ability to install (or publish) and update a software system (or library)
- Designing for deployability may involve:
 - Packaging (e.g., as a JAR file, Docker image)
 - Installer program
 - Checking for updates (e.g., automatic, requested by user, through app store)
 - Downtime and reinitialization for updates
 - Runtime environment requirements verification
 - Installing and updating plug-ins and add-ons
 - Backward compatibility



Questions?

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